LAB EXPERIMENT - 6

EXPERIMENT 6: WRITE A FORENSIC REPORT BASED ON SIMULATED ARTIFACTS

Aim

To analyze log files and produce a formal forensic report.

Inputs

- Log files
- System artifacts

Tools Used

- Log viewer (Notepad)
- Text editor

Dataset Required

- Authentication and access logs

➢ **Brief Theory**

Digital forensic reports are used to document evidence in a structured and legally acceptable manner. Log analysis helps in identifying suspicious activities such as unauthorized access, brute force attacks, and injection attempts.

➢ **Procedure**

Step 1: Create/Collect Log File

A sample log file was created using a text editor containing system activity records such as login attempts and file access.

Step 2: Open Log File

The log file was opened using Notepad for analysis.

Step 3: Analyze Logs

The logs were carefully examined to identify unusual patterns such as:

- Multiple failed login attempts
- Unauthorized file access
- Suspicious query execution

Step 4: Identify Suspicious Activities

The following threats were identified:

- Brute force attack
- Unauthorized access
- SQL Injection

Step 5: Extract Evidence

Relevant log entries were selected as evidence for the report.

Step 6: Prepare Forensic Report

A structured report was created including findings, evidence, and conclusions.

- **Sample Log File (Evidence)**

log.txt

File Edit View

```
2026-04-10 10:00:23 LOGIN SUCCESS user=admin ip=192.168.1.5
2026-04-10 10:02:10 LOGIN FAILED user=admin ip=192.168.1.10
2026-04-10 10:02:12 LOGIN FAILED user=admin ip=192.168.1.10
2026-04-10 10:02:14 LOGIN FAILED user=admin ip=192.168.1.10
2026-04-10 10:03:01 LOGIN FAILED user=admin ip=192.168.1.10
2026-04-10 10:05:00 ACCESS FILE confidential.txt user=guest
2026-04-10 10:06:45 SQL ERROR query=' OR 1=1 --
2026-04-10 10:07:12 LOGIN SUCCESS user=admin ip=192.168.1.10
```

- **Analysis & Findings**

1. Brute Force Attack

Multiple failed login attempts from IP **192.168.1.10** indicate a brute force attack.

2. Unauthorized Access

User **guest** accessed a confidential file, which indicates unauthorized access.

3. SQL Injection Attempt

The query ' OR 1=1 -- indicates a SQL injection attack.

- **Evidence Table**

Timestamp	Activity	Suspicion
10:02:10	Multiple login failures	Brute force attack
10:05:00	File access by guest	Unauthorized access
10:06:45	SQL error query	SQL Injection

- **Result**

Suspicious activities such as brute force attack, unauthorized access, and SQL injection were successfully identified through log analysis.

- **Conclusion**

Log analysis plays a crucial role in digital forensics by helping identify security threats and providing evidence for investigation and reporting.